

Weekly Diary – Week 2

Internship Organization: Vikaspedia

Duration: 24/01/2026 – 31/01/2026

Objective of the Week

The goal of Week 2 was to develop real-time synchronized text highlighting alongside Text-to-Speech playback, address synchronization and timing challenges, perform extensive cross-browser and mobile testing, and complete the project with proper documentation and final submission.

Day 1

- Studied various techniques for synchronizing spoken audio with text elements in the DOM.
- Explored the use of onboundary events and alternative timing-based synchronization methods.
- Compared word-level and sentence-level highlighting approaches.
- Selected the most reliable strategy based on browser compatibility and performance.

Day 2

- Implemented logic to break text content into individual words and wrap them within span elements.
- Linked spoken text segments with their respective DOM elements.
- Applied dynamic text highlighting using CSS classes and background styling.
- Ensured that only the currently spoken word or sentence remains highlighted.

Day 3

- Debugged synchronization inconsistencies caused by variations in speech rate.
- Ensured smooth and accurate highlight progression during pause and resume operations.
- Added logic to clear highlights when speech playback is stopped or completed.
- Enhanced word-tracking accuracy for longer text passages.

Day 4

- Performed cross-browser testing on Chrome, Firefox, Edge, and Opera.
- Analyzed differences in TTS event behavior and text highlighting across browsers.
- Identified limitations in certain browsers related to word boundary event support.
- Documented all findings for inclusion in the project README.

Day 5

- Tested the application on mobile browsers across Android and iOS platforms.
- Confirmed that the user interface remains clear and accessible on smaller screens.
- Verified that text highlighting stays synchronized with audio playback on mobile devices.

- Resolved minor layout and alignment issues to improve mobile usability.

Day 6

- Refactored the codebase to enhance readability, modular design, and maintainability.
- Optimized state management and improved component structure.
- Enhanced UI and UX elements with a focus on accessibility and user clarity.
- Removed unused variables and redundant logic to streamline the code.

Day 7

- Created a detailed README file covering:
 - Project overview
 - Installation and setup instructions
 - Supported languages and browsers
 - Known limitations and testing observations
- Uploaded the finalized project to a public GitHub repository.
- Conducted a final review to ensure all internship submission requirements were met.

Conclusion

- Successfully implemented synchronized text highlighting integrated with TTS playback.
- Completed thorough testing across multiple browsers and mobile devices.
- Delivered a multilingual, accessibility-focused frontend solution.
- Finalized and documented the project in accordance with Vikaspedia internship guidelines.